

Introduction to Program Synthesis (SS 2026)

Problem Domains

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Advanced Methodologies

Problem Domains: Symbolic Regression (SR)

- ▶ Search for mathematical expressions ¹
 - ↪ Represents a regression model that accurately and compactly describes the data
- ▶ Given a set of m points $\{x_i, y_i\}_{i=1\dots m}$, the task is to search for an expression that represents the function $f(x) \approx y$
- ▶ SR is a NP-hard problem
- ▶ Polynomials benchmarks → Koza-1, Koza-2 and Koza-3

Advanced Methodologies

Problem Domains: Logic Synthesis

- ▶ Logic synthesis → black-box optimisation task ²
- ▶ Synthesis of Boolean expressions → formal way of expressing Boolean functions
- ▶ **Hamming distance** → common loss function choice
 - ↪ Measures the distance between the candidate and target output values bitwise
 - ↪ Minimizing objective → distance of zero is ideal
- ▶ **Majority function** → Boolean function used for benchmarking
 - ↪ Returns true when half or more arguments are true and false otherwise

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Problem Domains: Logic Synthesis

x_1	x_2	x_3	y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Table: Truth table for the 3-bit majority function

Advanced Methodologies

Problem Domains: Program Synthesis

- ▶ Discovery of a program that matches the specification in the dataset $\mathcal{D}^3$
 - ↪ Set of positive and negative examples
- ▶ Distance is measured by the number of cases that predicted correctly by the candidate
 - ↪ Ideal fitness is then $|\mathcal{D}|$
 - ↪ Maximizing objective
- ▶ Learning by example or demonstration

Advanced Methodologies

Problem Domains: Policy Search

- ▶ Policy learning $\rightarrow$ discovery of decision-making strategies that maximise performance through interaction with an environment
 - $\leadsto$ Agent interacts with the environment and observes rewards
- ▶ The environment is formalised as a **Markov Decision Process (MDP)**
 - $\leadsto$ Defined by a 4-tuple (S, A, P, R) representing states, actions, transition probabilities and rewards
- ▶ The policy π interacts with the MDP to collect episodes $\tau = (s_0, a_0, r_1, s_1, \dots, s_T)$, i.e. sequences of states s , actions a and rewards r over time steps $t = 0, \dots, T$.
 - $\leadsto$ The fitness of a policy is measured as the expected accumulated reward
 - $\leadsto$ Maximizing objective $\rightarrow$ maximize reward